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BOTANY

1. Match List I with List II.

List I	List II
A) 40% lipids and 52% proteins	I) Mitochondria
B) Membrane with electron carriers	II) Chloroplast
C) Membrane structures in stacks	III) Plasma membrane
D) No membrane covering	IV) Ribosomes

Select the correct option.

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-II, C-IV, D-III

2. Which of the following hormones breaks seed dormancy, initiates germination in peanut seeds, and forms apical hook in dicots?

(1) Auxin

(2) Cytokinin

(3) Ethylene

(4) ABA

3. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): Respiration is an enzyme mediated amphibolic pathway that occurs in cytoplasm and mitochondria.

Reason (R): Respiration is mainly an anabolic process, but the intermediates of this anabolic process are utilised as substrates for a catabolic process too.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) (A) is false, but (R) is true.

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

4. Match List I with List II.

List I	List II
A) Golgi body	I) Protein synthesis
B) Ribosome	II) Formation of glycoprotein
C) Lysosome	III) Hydrolysis of macro molecules
D) Mitochondria	IV) Oxidation of food

Select the correct option.

(1) A-II, B-I, C-IV, D-III

(2) A-II, B-I, C-III, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-I, B-II, C-IV, D-III

5. The linking of antibiotic resistance gene with the plasmid vector is achieved using

(1) DNA polymerase

(2) restriction endonuclease

(3) *Taq* DNA polymerase

(4) DNA ligase

6. Match List I with List II.

List I	List II
A) Calyx in the form of pappus	I) Malvaceae
B) Parietal placentation	II) Asteraceae
C) Flower with epicalyx	III) Poaceae
D) Caryopsis fruit	IV) Brassicaceae

Select the correct option.

(1) A-II, B-I, C-IV, D-III

(2) A-II, B-I, C-III, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-II, C-IV, D-III

7. The margins of sepals or petals overlap one another but not in any particular direction in the flowers of

(1) pea, bean, and cotton

(2) *Cassia* and gulmohar

(3) china rose and lady's finger

(4) *Calotropis*, cotton, and gulmohar

8. Which of the following pairs is incorrectly matched?

(1) Agar-agar – Red algae

(2) Algin – Brown algae

(3) Carrageen – Red algae

(4) Laminarin – Green algae

9. If a tissue consumes 50 molecules of O_2 and releases 45 molecules of CO_2, to which category would the substrate utilised in cellular respiration belong to?

(1) Fats

(2) Simple sugars

(3) Proteins

(4) Polysaccharides

10. Given below are two statements.

Statement I : Photorespiration takes place in maize and bell pepper plants.

Statement II : The Hatch – Slack pathway occurs in sugarcane and tomato plants.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

11. Given below are two statements.

Statement I : *Lactobacillus* and *Penicillium* are used to obtain household products.

Statement II : *Saccharomyces* and *Trichoderma* are extensively used in medicine for the production of bioactive compounds.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

12. Which of the following is **not** related to prions?
    (1) Proteinaceous infectious agents
    (2) Infectious low molecular weight RNA
    (3) Causative agent of mad cow disease
    (4) Devoid of genetic material

13. Seeds in black pepper are perispermic due to persistent
    (1) endosperm
    (2) pericarp
    (3) nucellus
    (4) perianth

14. Read the following statements (A to D).
    A. Animals, mammals, and dogs represent taxa at different levels.
    B. Biological names are generally derived from Greek.
    C. A group of related genera comprises a family.
    D. In ascending hierarchy, the difficulty of classification decreases.

    Which of the following statements are correct?
    (1) A and B
    (2) B and C
    (3) A and C
    (4) C and D

15. In eukaryotes, RNA polymerase I is employed in the production of
    (1) rRNA (28S, 18S, and 5.8 S)
    (2) t-RNA, 5S r-RNA, Sn RNA
    (3) precursor of m-RNA
    (4) only Sn RNA

16. Read the following statements carefully and find out the **incorrect** statements.
    A. The capacity to generate a whole plant from any cell/explant is called tissue culture.
    B. The Green Revolution succeeded in tripling the food supply only due to the use of improved crop varieties.
    C. The proteins encoded by the genes *cry*IAc and *cry*IAb control the cotton bollworms and corn borer respectively.
    D. In RNAi technology, the source of complementary RNA could be from mobile genetic elements.
    (1) A and B
    (2) C and D
    (3) B and C
    (4) A and D

17. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).
    Assertion (A): RNA is the most efficient material for transmitting genetic information within the cell.
    Reason (R): RNA can directly code for synthesis of proteins.
    In light of the above statements, choose the correct answer from the options given below.
    (1) (A) is true, but (R) is false.
    (2) (A) is false, but (R) is true.
    (3) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
    (4) Both (A) and (R) are true, and (R) is the correct explanation of (A).

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18. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): During photosynthesis, CO_2 fixation occurs twice in C_4 plants.

Reason (R): First CO_2 fixation occurs in mesophyll cells with phosphoenol pyruvate, and second CO_2 fixation occurs in mesophyll cells with RuBP.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) (A) is false, but (R) is true.

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

19. A modified stem helps in protection in **A** while in **B**, it aids in perennation. Identify **A** and **B** and choose the correct option.

(1) A – *Cassia*
B – Turnip

(2) A – *Calotropis*
B – Tobacco

(3) A – Cotton
B – Tomato

(4) A – *Bougainvillea*
B – Turmeric

20. Given below are two statements.

Statement I : Hypodermis in monocot stem is a dead tissue, while in dicot stem, it is a living tissue.

Statement II: Phloem parenchyma is absent in gymnosperms, while sieve tubes are absent in monocots.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

21. Match List I with List II w.r.t. the structure of proteins.

	List I		**List II**
A)	Adult haemoglobin	I)	C-terminal
B)	Last amino acid	II)	Biologically active form
C)	Tertiary structure	III)	Quaternary structure
D)	Secondary structure	IV)	Right-handed helix

Select the correct option.

(1) A-I, B-III, C-II, D-IV

(2) A-III, B-I, C-II, D-IV

(3) A-IV, B-II, C-III, D-I

(4) A-II, B-III, C-I, D-IV

22. Match List I with List II.

List I	List II
A) Sticky pollen grains	I) *Vallisneria*
B) Pollen allergy	II) Anemophily
C) Light and dry pollen grains	III) *Parthenium*
D) Pollen covered by mucilage	IV) Entomophily

Select the correct option.

(1) A-II, B-IV, C-I, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-II, D-I

23. Which of the following organisms, techniques, and materials were used in the experiment for the proof of semi-conservative mode of DNA replication?

A. $CsCl_2$ density gradient centrifugation

B. Bacteriophage

C. *E.coli*

D. Radio isotopes of nitrogen

(1) A and B only

(2) A and C only

(3) C and D only

(4) B and C only

24. Read the following statements carefully and find out the **incorrect** statements.

A. A native plasmid is autonomously, replicating, circular extra chromosomal DNA.

B. Use of genetic engineering allows us to isolate and introduce only a set of desirable genes without introducing undesirable genes into the target organism.

C. DNA polymerases are also known as molecular scissors.

D. *Bam* HI and *Sal* I are selectable markers.

E. Micro-injection and biolistic methods are vector mediated methods to introduce alien DNA into host cells.

(1) A, B, and C

(2) B, C, and E

(3) C, D, and E

(4) B, C, and D

25. Which one of the following statements about histone proteins is correct?

(1) The nucleosome core is organised by a unit of 10 histone molecules.

(2) The pH of histone proteins is slightly alkaline.

(3) Histones are rich in amino acids lysine and tryptophan.

(4) Histones carry an overall negative charge in their side chains.

26. A pure tall plant with small starch grain in the seeds (TTbb) is crossed with a pure dwarf plant with large starch grain in the seeds (ttBB) and the plants of F_1 generation are self-fertilised. Find out the possible phenotype ratio in the F_2 generation.

(1) 1:2:1:2 (2) 9:3:3:1

(3) 3:6:3:1:2:1 (4) 1:2:3:1

27. Which of the following phenomena do the Human ABO blood groups represent?

(1) Polygenic inheritance

(2) Incomplete dominance

(3) Multiple allelism

(4) Linkage

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28. Identify the **incorrect** combination of technique and its purpose with respect to rDNA technology.
(1) Gel electrophoresis – Size-based separation of DNA fragments
(2) Polymerase chain reaction – Multiplication of desired gene *in vitro*
(3) Competent cell – able to take up naked DNA from surrounding medium
(4) Continuous cell culture – Maintain cell culture in lag phase

29. Which of the following contain haploid structures in an embryo sac?
(1) Nucellus and antipodal
(2) Antipodal and synergid
(3) Antipodal and megaspore mother cell
(4) Nucellus and generative cell

30. Match List I with List II.

	List I		List II
A)	Leptotene	I)	Appearance of recombination nodules
B)	Zygotene	II)	First stage of prophase-I
C)	Pachytene	III)	Transition to metaphase
D)	Diakinesis	IV)	Pairing of homologous chromosomes

Select the correct option.
(1) A-II, B-IV, C-I, D-III
(2) A-II, B-III, C-IV, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-I, C-II, D-III

31. Given below are two statements.

Statement I : In some leguminous plants, leaf base is termed pulvinus.

Statement II : In silk cotton, leaves are pinnately compound.

In light of the above statements, choose the correct answer from the options given below.
(1) Statement I is correct, but statement II is incorrect.
(2) Both statement I and statement II are correct.
(3) Both statement I and statement II are incorrect.
(4) Statement I is incorrect, but statement II is correct.

32. Identify the correct combination.
(1) Insulin – Enzyme
(2) Protein – Homopolymer
(3) Glut-4 – Enables glucose transport
(4) Collagen – Intracellular substance

33. The site for photosynthesis in a prokaryotic cell is
(1) gas vacuole
(2) ribosome
(3) chromatophore
(4) polysome

34. After digesting a DNA sample with a restriction enzyme, the sizes of fragments obtained are as follows:

P = 5 kb; Q = 3 kb; R = 8 kb; S = 10 kb

What would be the expected sequence of bands from anode to cathode after subjecting this mixture of fragments to gel electrophoresis?
(1) P, Q, R, and S (2) Q, P, R, and S
(3) S, R, Q, and P (4) R, P, Q, and S

35. Match List I with List II.

List I	List II
A. Insecticidal protein	I) Silencing of gene expression
B. Nematode infestation	II) Virus free plants
C. Pomato	III) Becomes activated at alkaline pH
D. Axillary meristem	IV) Protoplasmic fusion

Select the correct option.

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-III, C-II, D-I

36. Given below are two statements.

Statement I : Interkinesis is a short phase between the interphase and M phase of cell cycle.

Statement II : Megasporogenesis and microsporogenesis involve meiotic divisions.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

37. Which of the following statements are correct?

A. Phaeophyceae members have chlorophyll *a* and *c*.

B. *Selaginella* has the condition referred to as precursor of seed habit.

C. Pinus has needle shaped leaves.

D. Mosses along with ferns are the first organisms to colonise rocks.

Choose the correct answer from the options given below.

(1) A, B, and D only

(2) B, C, and D only

(3) A, C, and D only

(4) A, B, and C only

38. Match List I with List II. w.r.t. F_2 generation of Mendel's dihybrid cross.

List I (Genotype)	List II (Probability)
A) RRYy	I) 6.25%
B) RrYy	II) 25%
C) rryy	III) 6.25%
D) rrYY	IV) 12.5%

Select the correct option.

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-II, C-III, D-I

(4) A-I, B-II, C-III, D-IV

39. Which of the following non-membranous cell organelle is made up of non-genetic nucleic acid and proteins?

(1) Chloroplast

(2) Ribosome

(3) Centrosome

(4) Mitochondrion

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40. Match List I with List II.

List I	List II
A) S phase	I) Reorganisation of cellular components
B) G_1 phase	II) Centriole duplication
C) G_0 phase	III) Increase in the cytoplasmic volume
D) M phase	V) No cell proliferation

Select the correct option.

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-IV, B-III, C-II, D-I

41. Given below are two statements.

Statement I : In arithmetic growth, all the daughter cells formed in every generation divide further to produce daughter cells.

Statement II : A sigmoid growth curve is a characteristic of living organism growing in a natural environments.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Statement I is correct, but statement II is incorrect.

42. Match List I with List II.

List I	List II
A) Ribozyme	I) Structural polysaccharide
B) Cellulose	II) Organic cofactor
C) Antibiotic	III) Catalytic nucleic acid
D) Prosthetic group	IV) Competitive inhibitor

Select the correct option.

(1) A-IV, B-III, C-II, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-III, D-IV

43. A monocot root differs from a monocot stem in having

(1) open vascular bundles having endarch xylem

(2) scattered vascular bundles having protoxylem towards periphery

(3) closed vascular bundles with protoxylem towards the centre

(4) radial vascular bundles with exarch xylem

44. In Ascomycetes and Basidiomycetes, the intervening period between plasmogamy and karyogamy is

(1) diplophase

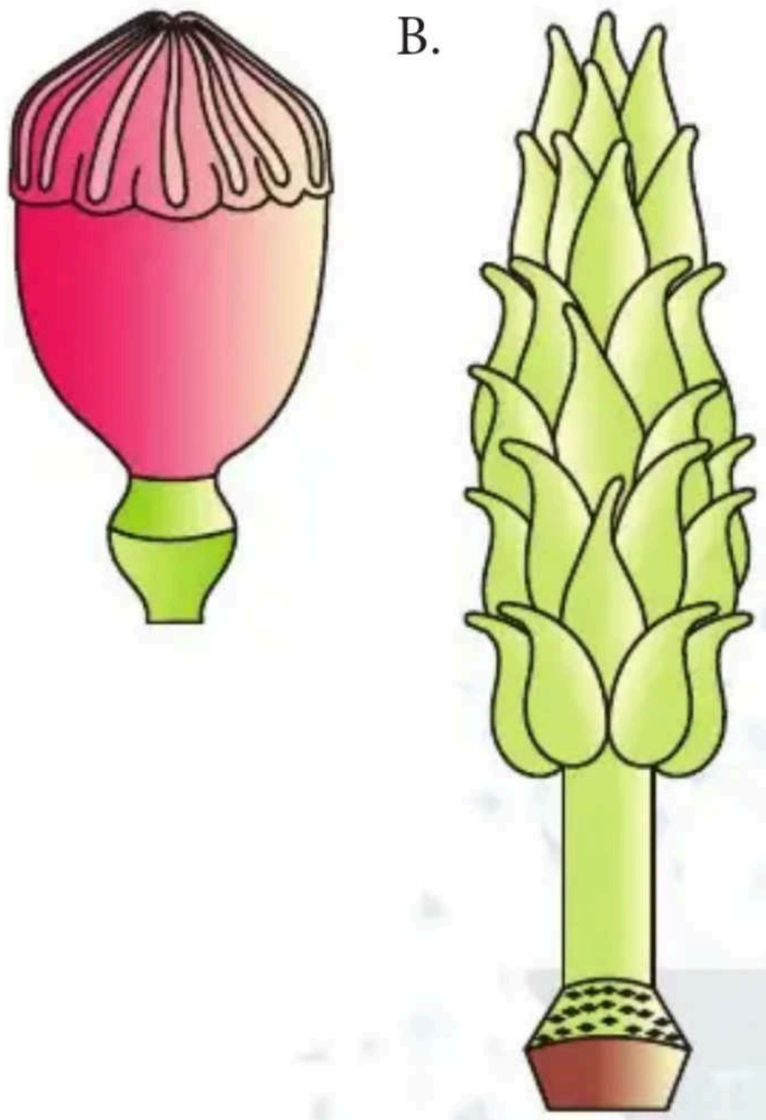
(2) dikaryophase

(3) interphase

(4) resting phase

45. Observe the diagrams and identify the correct option that represents the condition and corresponding examples.

A. B.



(1) A – Syncarpous – *Papaver* and
B – Apocarpous – *Michelia*

(2) A – Apocarpous – *Michelia* and
B – Syncarpous – *Papaver*

(3) A – Hypogynous – Sunflower and
B – Epigynous – Chinarose

(4) A – Perigynous – *Datura* and
B – Hypogynous – Peach

ZOOLOGY

46. The Rivet Popper Hypothesis compares an aeroplane to **A** and the rivets to **B**. Select the option that correctly identifies A and B.

(1) A – Ecosystem
B – Species

(2) A – Community
B – Population

(3) A – Population
B – Individuals

(4) A – Ecosystem
B – Community

47. Match List I with List II.

List I	**List II**
A) Oviparous mammal	I) *Branchiostoma*
B) Freshwater sponge	II) *Struthio*
C) Flightless bird	III) *Spongilla*
D) Invertebrate chordate	IV) *Ornithorhynchus*

Select the correct option.

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-IV, C-I, D-II

48. Which of the following is a component of the logistic growth equation but not of the exponential growth equation?

(1) N (2) r

(3) K (4) t

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49. A genetic disorder is characterised by the following features:

- Rudimentary ovaries
- Short stature and a webbed neck
- Absence of secondary sexual characteristics
- Broad chest with widely spaced nipples
- Normal intelligence but possible learning difficulties in some cases
- The individual is a female, with a karyotype of 45, XO

Which of the following conditions matches the description given above?

(1) Klinefelter's syndrome

(2) Turner's syndrome

(3) Down's syndrome

(4) Edward's syndrome

50. Select the **incorrect** one from the following statements related to human reproduction.

(1) Spermiogenesis is the process by which spermatids are transformed into spermatozoa.

(2) Spermiation is the release of spermatozoa into the lumen of the seminiferous tubules.

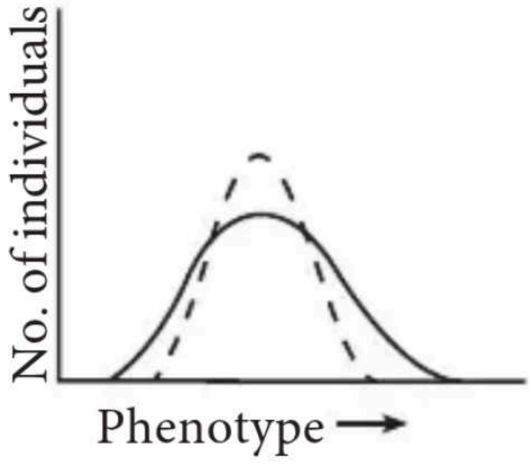
(3) Oogenesis is initiated during the embryonic development stage itself.

(4) Ovulation is the release of the ovum from the ovary following the completion of oogenesis.

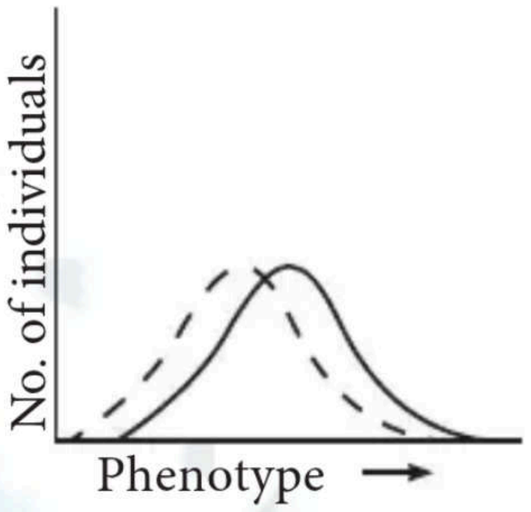
51. Which of the following diagrams best represents stabilising selection?

Note: Dashed lines (— — —) represent the original population and solid lines (————) represent the evolved population.

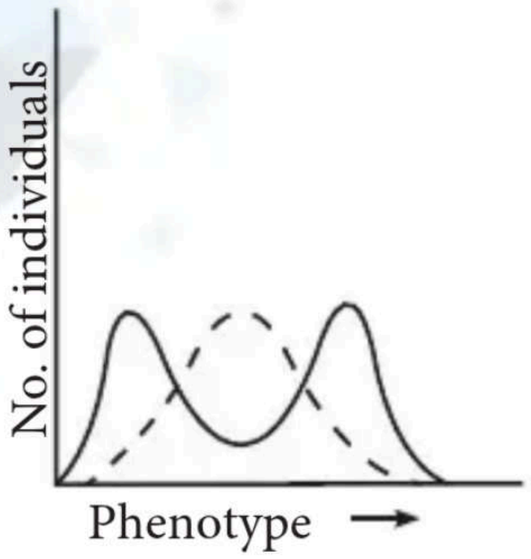
(1)



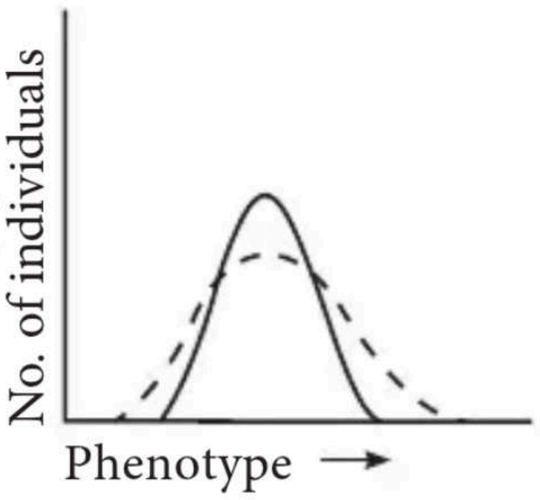
(2)



(3)



(4)



52. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): If the head of a cockroach is cut off, it will still live for as long as one week.

Reason (R): The entire nervous system is situated along the ventral (belly-side) part of the thorax and abdomen.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

53. The cellular barrier of innate immunity includes all the following, except

(1) neutrophils

(2) monocytes

(3) macrophages

(4) basophils

54. In which of the following ecosystems does the first trophic level have more organisms than the second trophic level, while the second trophic level has more biomass than the first trophic level?

(1) Forest ecosystem

(2) Grassland ecosystem

(3) Tree ecosystem

(4) Ocean ecosystem

55. The cells lining the air sacs of the lungs comprise

(1) squamous epithelium

(2) cuboidal epithelium

(3) columnar epithelium

(4) compound epithelium

56. The structural and functional unit of the vertebrate kidney is

(1) Malpighian tubule

(2) renal pyramid

(3) uriniferous tubule

(4) renal corpuscle

57. Match List I with List II.

List I	List II
A) Superior vena cava	I) Carries deoxygenated blood to lungs
B) Inferior vena cava	II) Carries oxygenated blood from lungs
C) Pulmonary artery	III) Brings deoxygenated blood from lower parts of the body to the right atrium
D) Pulmonary vein	IV) Brings deoxygenated blood from upper parts of the body to the right atrium

Select the correct option.

(1) A-II, B-IV, C-III, D-I

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-I, C-III, D-II

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58. Given below are two statements.

Statement I : The principle of vaccination is based on the memory of the immune system.

Statement II : ELISA is based on the principle of antigen-antibody interaction.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

59. Given below are two statements.

Statement I : Neurons are excitable cells because their membranes are in a polarised state.

Statement II : When a neuron is not conducting any impulse, i.e., resting, the inner surface of the axonal membrane possesses a positive charge while its outer surface is negatively charged.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct but statement II is incorrect.

(3) Both statement I and statement II are incorrect.

(4) Statement I is incorrect but statement II is correct.

60. The partial pressures of oxygen (pO_2) and carbon dioxide (pCO_2) at two sites are given below.

	pO_2	**pCO_2**
A	95 mm Hg	40 mm Hg
B	40 mm Hg	45 mm Hg

Select the option that correctly identifies **A** and **B**.

(1) A – Pulmonary artery,
B – Pulmonary vein

(2) A – Pulmonary artery,
B – Vena cava

(3) A – Pulmonary vein,
B – Pulmonary artery

(4) A – Alveolus,
B – Dorsal aorta

61. Select the **incorrect** statement from the following.

(1) The bone marrow in all bones of adult humans is the site of blood cell production.

(2) Most of the cartilages in vertebrate embryos are replaced by bones in adults.

(3) Fibroblasts secrete modified polysaccharides, which accumulate between cells of connective tissue.

(4) Simple epithelium functions as a lining for body cavities, ducts, and tubes.

62. In the case of industrial melanism, the rise in the frequency of dark-coloured peppered moths in industrial areas of England was primarily caused by the
(1) mutations induced by soot from industries
(2) immigration of dark-coloured moths in large numbers
(3) extinction of lichens due to predation by moths
(4) selection of pre-existing mutation

63. The total biomass of living organisms present in a given area at a specific time is termed
(1) gross production
(2) gross productivity
(3) standing crop
(4) standing state

64. Match List I with List II.

	List I		List II
A)	Tarsals	I)	Pectoral girdle
B)	Carpals	II)	Pelvic girdle
C)	Glenoid cavity	III)	Ankle bones
D)	Acetabulum	IV)	Wrist bones

Select the correct option.
(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-I, D-II

65. The table given below represents the outcomes of interspecific interactions between different species where:
'+' indicates a beneficial interaction
'–' indicates a detrimental interaction
'0' indicates a neutral interaction

Species 2 \ Species 1	+	–
+	A	B
–	C	D
0	E	F

Select the option that correctly identifies two of the six interactions (A–F).
(1) A – Mutualism
B – Competition
(2) C – Predation
D – Parasitism
(3) E – Commensalism
F – Amensalism
(4) C – Amensalism
D – Competition

66. Match List I with List II.

	List I		List II
A)	Inspiratory capacity	I)	4000 to 4600 mL
B)	Expiratory capacity	II)	2100 to 2300 mL
C)	Functional residual capacity	III)	1500 to 1600 mL
D)	Vital capacity	IV)	3000 to 3500 mL

Select the correct option.
(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-I, C-IV, D-III

Infinity Learn

67. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): Cancer cells continue to divide giving rise to masses of cells called tumours.

Reason (R): Cancer cells have lost the property of contact inhibition.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

68. Given below are two statements.

Statement I : Somatostatin inhibits the release of growth hormone from the pituitary gland.

Statement II : GnRH stimulates the release of gonadotropins (FSH and LH) from the gonads.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct but statement II is incorrect.

(3) Both statement I and statement II are incorrect.

(4) Statement I is incorrect but statement II is correct.

69. Select the **incorrect** one from the following statements about human development.

(1) The cleavage starts as the zygote moves through the isthmus of the oviduct towards the uterus.

(2) The chorionic villi and uterine tissue become interdigitated with each other and jointly form the placenta.

(3) Immediately after implantation, the trophoblast differentiates into ectoderm and endoderm.

(4) The inner cell mass contains certain stem cells that have the potency to give rise to all the tissues and organs.

70. Which of the following are **not** organised endocrine bodies in the human body but produce hormones?

A. Gastrointestinal tract

B. Liver

C. Heart

D. Thyroid

E. Kidney

(1) A, C, and E only

(2) B, C, and E only

(3) B and D only

(4) A, B, C, and E

71. The gizzard of a cockroach helps in grinding food particles since it contains a thick layer of

(1) circular muscles and chitinous teeth

(2) longitudinal muscles and chitinous teeth

(3) circular muscles and hepatic caeca

(4) outer cuticle and inner chitinous teeth

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72. Match List I with List II.

List I (ECG component)	**List II (Electrical activity)**
A) P wave	I) Atria – resting, Ventricles – repolarisation
B) QRS complex	II) Atria – depolarisation, Ventricles – resting
C) T wave	III) Atria – repolarisation, Ventricles – depolarisation
D) TP segment	IV) Atria – resting, Ventricles – resting

Select the correct option.

(1) A-II, B-III, C-I, D-IV

(2) A-IV, B-II, C-I, D-III

(3) A-I, B-III, C-IV, D-II

(4) A-III, B-II, C-IV, D-I

73. Which of the following outcomes may result from Addison's disease?

A. Decreased blood pressure

B. Increased protein breakdown

C. Elevated blood glucose levels

D. Decreased Na^+ levels

Which of the following options is the most appropriate?

(1) A and B are correct.

(2) A and C are correct.

(3) B and D are correct.

(4) A and D are correct.

74. Match List I with List II.

List I	**List II**
A) Conditional reabsorption of Na^+ and water	I) DCT
B) Active reabsorption of glucose and amino acids	II) Collecting duct
C) Passive reabsorption of urea	III) Ascending limb of Henle's loop
D) Minimal reabsorption	IV) PCT

Select the correct option.

(1) A-III, B-II, C-IV, D-I

(2) A-I, B-IV, C-II, D-III

(3) A-IV, B-I, C-II, D-III

(4) A-I, B-IV, C-III, D-II

75. A man with normal vision, whose father was colour-blind, marries a woman with normal vision, whose father was also colour-blind. What is the probability of their daughter being colour-blind?

(1) 1

(2) 0.5

(3) 0.25

(4) 0

76. Which of the following animals does **not** exhibit sexual dimorphism?

(1) *Ascaris*

(2) *Periplaneta*

(3) *Rana*

(4) *Asterias*

77. Match List I with List II.

List I	List II
A) Contraction of myometrium	I) Low progesterone levels
B) Relaxation of myometrium	II) High progesterone levels
C) Proliferation of endometrium	III) Oxytocin
D) Shedding of endometrium	IV) Oestrogen

Select the correct option.

(1) A-III, B-II, C-IV, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-II, C-III, D-I

(4) A-I, B-IV, C-II, D-III

78. To the global species diversity, India contributes

(1) 6.7% (2) 22%

(3) 2.4% (4) 8.1%

79. Read the following statement with two blanks (A and B).

The urethra in male humans originates from the __(A)__ and extends through the __(B)__ to its external opening called the urethral meatus. The one correct option for the two blanks is

(1) A – ureter, B – penis

(2) A – urinary bladder, B – penis

(3) A – urinary bladder, B – scrotum

(4) A – epididymis, B – penis

80. Given below are two statements.

Statement I : Embryos from *in vivo* or *in vitro* fertilisation can be transferred into the uterus as part of assisted reproductive technology (ART).

Statement II : The collection and transfer of semen into the female reproductive tract is called ICSI.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

81. Which of the following statements are correct?

A. The cranial capacity of a Java man is more than that of *Homo habilis* but less than that of a Neanderthal man.

B. *Homo sapiens* was the first to practise the ritual of burying dead bodies.

C. The skull of a baby chimpanzee is more like an adult human skull than an adult chimpanzee skull.

D. Hominid evolution shows a dietary trend from herbivorous to omnivorous diets due to hunting and tool use.

Choose the correct answer from the options given below.

(1) A, B, C, and D

(2) A, B, and C only

(3) B, C, and D only

(4) A, C, and D only

82. The largest known gene in the human genome codes for

(1) trypsin (2) dystrophin

(3) myoglobin (4) myosin

83. Which of the following is a public hygiene measure that is very effective in preventing amoebiasis?

(1) Consumption of clean drinking water, food, vegetables, fruits, etc.

(2) Introducing fish that feed on mosquito larvae in ponds

(3) Avoiding close contact with infected persons

(4) Disinfection of water reservoirs, pools, cesspools and tanks

84. Match List I with List II.

List I	List II
A) Amoeboid movement	I) Oviducts
B) Ciliary movement	II) Limbs
C) Flagellar movement	III) Spermatozoa
D) Muscular movement	IV) Macrophages

Select the correct option.

(1) A-I, B-IV, C-III, D-II

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-I, C-III, D-II

(4) A-III, B-II, C-IV, D-I

85. Given below are two statements.

Statement I : The life cycles of endoparasites are more complex because of their extreme specialisation.

Statement II : The human liver fluke requires two hosts to complete its life cycle.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

86. The mesoderm is present as scattered pouches between the ectoderm and the endoderm in

(1) earthworm and silkworm

(2) tapeworm and roundworm

(3) tongue worm and roundworm

(4) hookworm and filarial worm

87. Match List I with List II.

List I	List II
A) Dorsal part of forebrain	I) Medulla oblongata
B) Ventral part of forebrain	II) Corpora quadrigemina
C) Dorsal part of midbrain	III) Pineal gland
D) Inferior part of hindbrain	IV) Pituitary gland

Select the correct option.

(1) A-IV, B-II, C-III, D-I

(2) A-III, B-IV, C-II, D-I

(3) A-I, B-II, C-IV, D-III

(4) A-II, B-I, C-III, D-IV

Infinity Learn

88. Match List I with List II.

List I	List II
A) Natural method	I) Vasectomy
B) Barrier method	II) Coitus interruptus
C) Hormonal method	III) Pills
D) Surgical method	IV) Diaphragms

Select the correct option.

(1) A-I, B-IV, C-III, D-II

(2) A-II, B-IV, C-III, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-IV, C-II, D-I

89. Given below are two statements.

Statement I : Fertilisation is internal in *Balanoglossus* and *Balaenoptera*.

Statement II : Development is indirect in *Asterias* and *Ascidia*.

In light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

90. Narrowly utilitarian, broadly utilitarian, and ethical arguments are the

(1) causes of biodiversity loss

(2) reasons for the conservation of biodiversity

(3) methods of biodiversity conservation

(4) levels of biodiversity

PHYSICS

91. A parallel plate capacitor of capacitance C has a charge q when a battery of emf V is connected across it. Now, the battery is removed. The work required to triple the distance between the plates is

(1) CV^2 (2) $\frac{1}{2}CV^2$

(3) $\frac{3}{2}CV^2$ (4) $\frac{1}{4}CV^2$

92. The current sensitivity of a galvanometer can be increased by

A. decreasing the number of turns

B. increasing the magnetic field

C. decreasing the area of the coil

D. decreasing the torsional constant of the spring

Choose the most appropriate answer from the options given below.

(1) B and C only

(2) C and D only

(3) A and C only

(4) B and D only

93. The instantaneous values of alternating current and voltages in a circuit are given as

$$i = \frac{1}{\sqrt{2}}\sin(100\pi t)\text{ ampere}$$

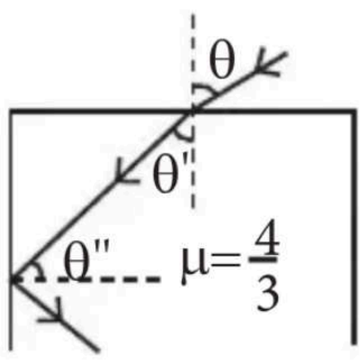
$$e = \frac{1}{\sqrt{2}}\sin\left(100\pi t + \frac{\pi}{3}\right)\text{ volt}$$

The average power in watts consumed in the circuit is

(1) $\frac{1}{4}$ (2) $\frac{\sqrt{3}}{4}$

(3) $\frac{1}{2}$ (4) $\frac{1}{8}$

94. A ray of light enters from air into a denser medium of refractive index $\frac{4}{3}$, as shown in the figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θ should be equal to



(1) $\sin^{-1}\frac{\sqrt{7}}{3}$ (2) $\sin^{-1}\frac{\sqrt{5}}{4}$

(3) $\sin^{-1}\frac{\sqrt{7}}{4}$ (4) $\sin^{-1}\frac{\sqrt{5}}{3}$

95. In a Young's double slit experiment, two slits are separated by 2 mm and the screen is placed one metre away. When a light of wavelength 500 nm is used, the fringe separation will be

(1) 1 mm (2) 0.25 mm

(3) 0.75 mm (4) 0.50 mm

96. The de-Broglie wavelength of a particle having kinetic energy E is λ. How much extra energy must be given to this particle so that the de-Broglie wavelength reduces to 75% of the initial value?

(1) $\frac{16}{9}E$ (2) E

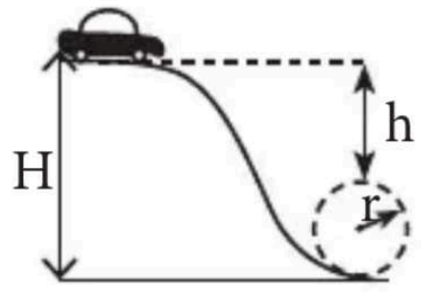
(3) $\frac{7}{9}E$ (4) $\frac{1}{9}E$

97. The radius of ${}^{x}_{32}\text{Ge}$ (germanium) nucleus is measured to be twice the radius of ${}^{9}_{4}\text{Be}$ nucleus. The number of neutrons in Ge are

(1) 38 (2) 40

(3) 42 (4) 64

98. A toy car slips down the smooth inclined plane as shown in the figure. It just goes around the vertical smooth circle at the bottom. The relation between H and h is:



(1) H = 2h (2) H = 3h

(3) H = 4h (4) H = 5h

99. If the tube length of an astronomical telescope is 105 cm and its magnifying power is 20 for normal setting, the focal length of the objective is

(1) 100 cm (2) 10 cm

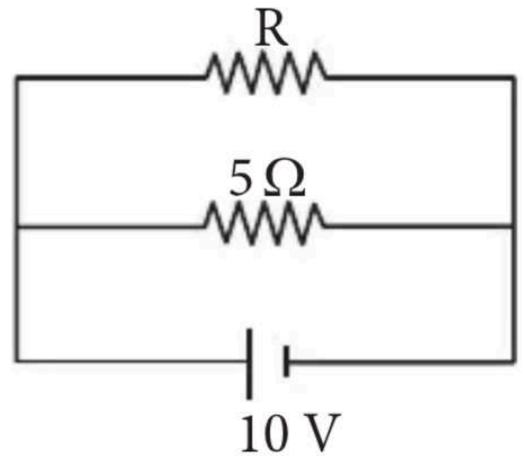
(3) 20 cm (4) 25 cm

100. In a pure silicon ($n_i = 10^{16}/m^3$) crystal at 300K, 10^{21} atoms of phosphorus are added per cubic meter. The new hole concentration will be

(1) 10^{21} per m^3 (2) 10^{19} per m^3

(3) 10^{11} per m^3 (4) 10^{5} per m^3

101. The power dissipated in the circuit shown in the figure is 30 watts. The value of R is



(1) 15 Ω (2) 10 Ω

(3) 30 Ω (4) 20 Ω

102. In equation $y = x^2 \cos^2 \frac{\beta\gamma}{\alpha}$, the dimensions of x, α, β are L, T^{-1} and $(LT^{-1})^{-1}$ respectively. The dimensions of y and γ are

(1) L^2, LT^{-2} (2) L, LT^{-1}

(3) L^2, L (4) L, LT^{-2}

103. A bus travelling at a speed of 30 m/s slows down with uniform retardation. It covers a distance of 90 m as its velocity reduces to 24 m/s. If the bus continues to slow down at the same rate, after what further distance will it be brought to stop?

(1) 24.5 m (2) 250 m

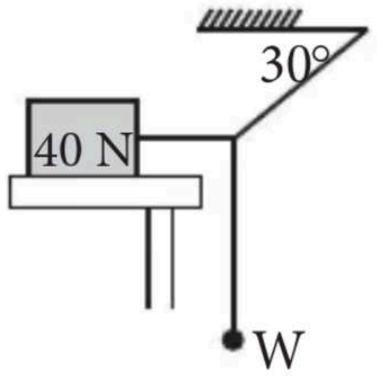
(3) 160 m (4) 200 m

104. The resultant of two forces is half of the magnitude of the larger force. The resultant is perpendicular to the smaller force. Then the angle between the forces is

(1) $\frac{\pi}{6}$ (2) $\frac{2\pi}{3}$

(3) $\frac{5\pi}{6}$ (4) $\frac{\pi}{3}$

105. In the given figure, the system is in equilibrium. What is the maximum value that W can have if the friction force on the block of weight 40 N cannot exceed 12.0 N?



(1) 3.45 N (2) 6.92 N

(3) 10.35 N (4) 12.32 N

106. In a hydroelectric power station, the water is flowing at 2 ms^{-1} in the river which is 100 m wide and 5 m deep. The average power output from the river is ($\rho_{water} = 10^3 kg/m^3$)

(1) 3 MW (2) 1.5 MW

(3) 2.5 MW (4) 2 MW

107. A disc of mass M and radius R is rotating about its axis. When a particle is gently placed at its rim, the angular speed becomes half. What is the mass of the particle? (Assume no slipping takes place between the particle and the disc.)

(1) $M/2$ (2) $M/3$

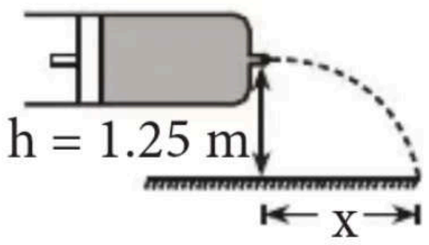
(3) $2M$ (4) $3M$

108. A metal wire A of length 5 m and area of cross section $2.5 \times 10^{-5} m^2$ is stretched uniformly by 2 mm under the application of a force F. Another metal wire of length 6 m and area of cross section $3 \times 10^{-5} m^2$ is stretched uniformly by 4 mm under the application of the same force F. The ratio of Young's moduli of A and B is

(1) 1 : 1 (2) 1 : 2

(3) 2 : 1 (4) 1 : 10

109. A tube has two different areas of cross section having diameters 8 mm and 2 mm as shown in the figure. The range of water falling on horizontal surface, if the piston is moving with a constant velocity of 0.25 ms^{-1}, is (h = 1.25 m and g = 10 ms^{-2})



(1) 1 m (2) 2 m

(3) 3 m (4) 4 m

110. An empty pressure cooker with a volume of 10 litres contains air at atmospheric pressure 10^5 Pa and temperature 27 °C. The cooker has a whistle with an area of 0.1 cm^2 and a mass of 100 grams.



What should be the temperature of air inside the pressure cooker so that the whistle is just lifted? (Neglect expansion of cooker) ($g = 10\ m/s^2$)

(1) 400 °C

(2) 227 °C

(3) 327 °C

(4) 600 °C

111. 1 kg of a diatomic ideal gas is at a pressure of $8 \times 10^4\ N/m^2$. The density of the gas is $4\ kg/m^3$. The energy of the gas due to its translational thermal motion will be

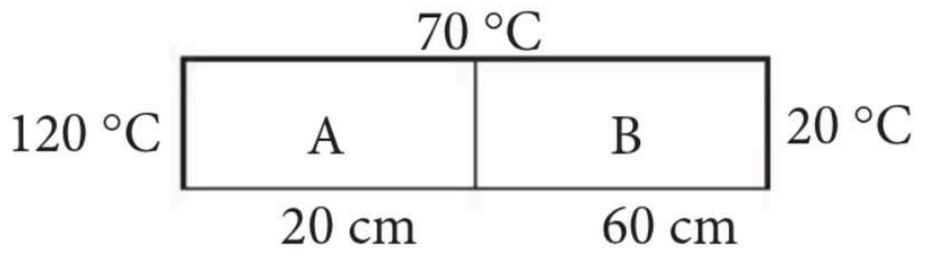
(1) 3×10^4 J

(2) 5×10^4 J

(3) 6×10^4 J

(4) 7×10^4 J

112. Two metal blocks A and B of the same cross-section area are connected to each other as shown in the figure. If the thermal conductivity of block A is $2k$, then the thermal conductivity of block B will be ____ (Assume the system is in steady state.)



(1) $3k$ (2) $5k$

(3) $6k$ (4) $k/3$

113. Given below are two statements.

Statement I : A second's pendulum has a time period of one second.

Statement II : A second's pendulum takes precisely one second to move between the two extreme positions.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are incorrect.

(2) Statement I is incorrect, but statement II is correct.

(3) Statement I is correct but statement II is incorrect.

(4) Both Statement I and statements II are correct.

114. In the experiment to determine speed of sound by a resonance tube, the difference between the first two resonance positions is

(1) equal to the wavelength of sound

(2) three fourth of the wavelength of sound

(3) half the wavelength of sound

(4) one-fourth of the wavelength of sound.

115. A short electric dipole is placed in the electric field due to a point charge. Then,

A. the dipole must experience a force

B. the dipole may experience a force

C. the dipole must experience a torque

D. the dipole may experience a torque

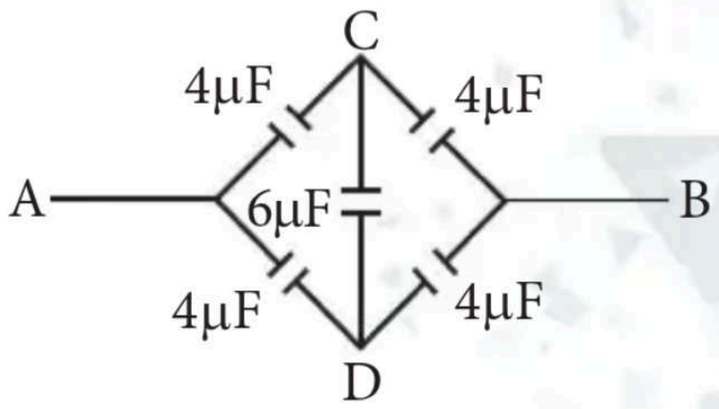
(1) A and D are correct

(2) B and D are correct

(3) C and D are correct

(4) D alone is correct

116. Find equivalent capacitance between C and D.



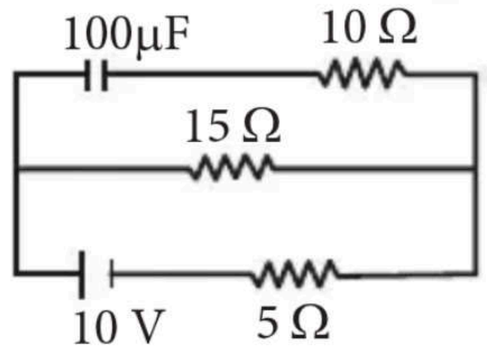
(1) 4 μF

(2) 8 μF

(3) 2 μF

(4) 10 μF

117. The charge stored on the capacitor shown in the circuit diagram at steady state is



(1) zero

(2) 250 μC

(3) 750 μC

(4) 1200 μC

118. A straight wire carrying current $2I$ is turned into a circular loop. If the magnitude of the magnetic moment associated with it is M, then the length of the wire will be:

(1) $\sqrt{\frac{4\pi M}{I}}$ (2) $\sqrt{\frac{2M\pi}{I}}$

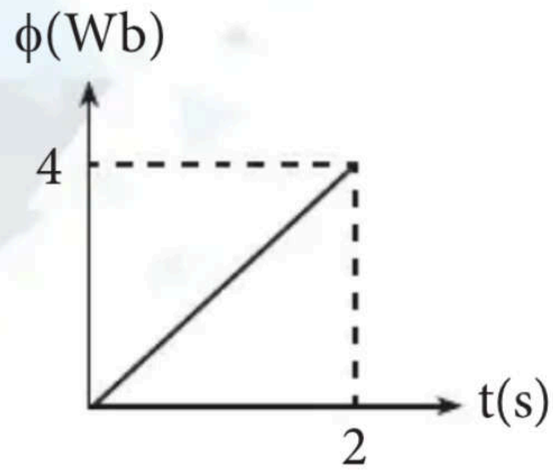
(3) $\sqrt{\frac{M\pi}{I}}$ (4) $\sqrt{\frac{3\pi}{MI}}$

119. A magnetic field strength (H) of magnitude 3×10^3 A/m produces a magnetic field of induction (B) of magnitude 12π tesla in an iron rod. The relative permeability of the iron is

(1) 10^5 (2) 10^4

(3) 10^3 (4) 10^2

120. The magnetic flux passing through a coil of resistance 0.5 Ω is as shown in the figure. Match the following two lists. In List II, all physical quantities are in SI units.



List I	List II
A) Induced emf produced	I) 4
B) Induced current	II) 16
C) Charge flow in 2s	III) 8
D) Heat generation in 2s	IV) 2

(1) A-IV, B-I, C-III, D-II

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-III, B-II, C-IV, D-I

121. An ideal transformer has 100 turns in the primary coil and carries 8 A current. If the input power is one kilowatt, the number of turns required in the secondary coil to have 500 V output will be

(1) 100 (2) 200

(3) 400 (4) 300

122. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): The small ozone layer on top of the stratosphere is crucial for human survival.

Reason (R): It absorbs ultraviolet radiation from the sun and prevents it from reaching the earth's surface and causing damage to life.

In light of the given statements, choose the correct answer from the options given below.

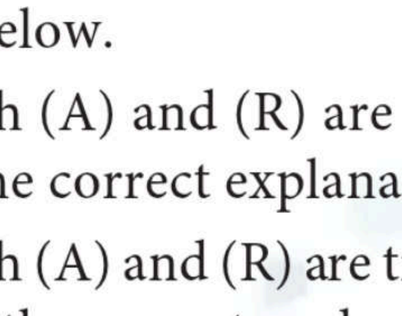
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

123. In the following figure, a coin at O is placed at the bottom of a pond and there are three observers, A, B, and C in the air.



(1) All the three observers will see the same image.

(2) The image seen by each observer will be at a different position.

(3) It is possible that any observer might not be able to see any image.

(4) Both (2) and (3) are true.

124. An un-polarised light of intensity I_0 is incident on surface of a block of glass at Brewster's angle. In this case, which one of the following statement is true?

(1) Transmitted light is partially polarised with intensity $\frac{I_0}{2}$.

(2) Transmitted light is completely polarised.

(3) Reflected light is completely polarised with intensity less than $\frac{I_0}{2}$.

(4) Reflected light is partially polarised with intensity $\frac{I_0}{2}$.

125. For atomic model of hydrogen atom given by Niels Bohr, match the following proportionalities.

List I	List II
A) Angular momentum	I) 1/n
B) Velocity of electron	II) n^2
C) Radius of orbit	III) $1/n^2$
D) Energy of electron	IV) n

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-IV, B-I, C-III, D-II

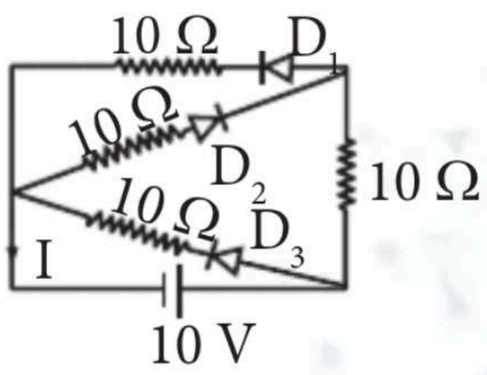
(4) A-IV, B-I, C-II, D-III

126. The binding energy of a certain nucleus is 18×10^8 J. How much is the difference between the total mass of all the free nucleons and the nuclear mass of the given nucleus?

(1) 20 μg (2) 2 μg

(3) 10 μg (4) 0.2 μg

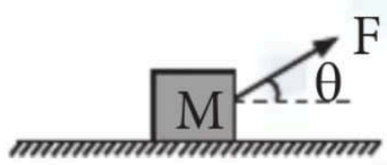
127. In the given circuit, the current (I) through the battery will be:



(1) 1A (2) 2.5 A

(3) 2A (4) 1.5 A

128. A block of mass M is pulled by a force F making angle θ with horizontal on a smooth horizontal surface as shown below. If a is the acceleration of the block on the surface, then the contact force between the block and the surface is



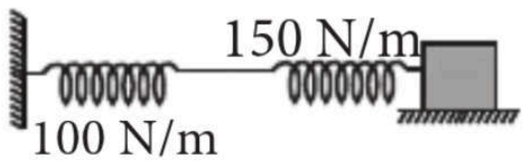
(1) $Mg + Ma\cos\theta$

(2) $Mg - Ma\cos\theta$

(3) $Mg + Ma\tan\theta$

(4) $Mg - Ma\tan\theta$

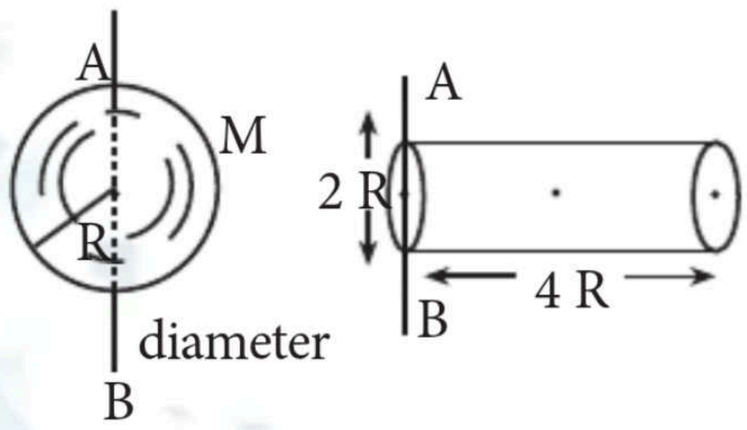
129. Two springs of force constant 100 N/m and 150 N/m are in series as shown. The block is pulled 2.5 cm to the right from the equilibrium position. What is the ratio of the work done by the left spring to the work done by the right spring?



(1) 3/2 (2) 2/3

(3) 0.2 (4) 2

130. The ratio of radius of gyration of a thin spherical shell to that of a solid cylinder of equal mass, for moment of inertia about their axis AB as shown in the figure, is $\sqrt{8/x}$. The value of x is



(1) 34 (2) 51

(3) 67 (4) 17

131. A particle of mass M is situated at the centre of a thin spherical shell of the same mass and radius a. The gravitational potential at a point situated at distance $\frac{a}{2}$ from the centre will be

(1) $-\frac{3GM}{a}$

(2) $-\frac{2GM}{a}$

(3) $-\frac{GM}{a}$

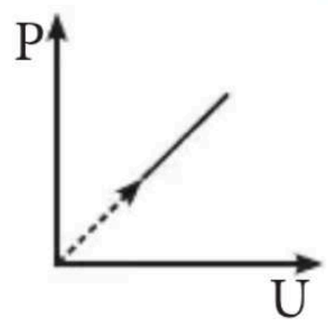
(4) $-\frac{4GM}{a}$

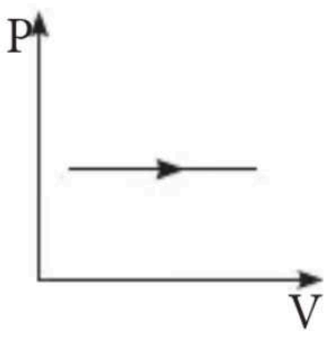
132. An object of mass 2 kg hanging with the help of a spring is in equilibrium condition. When this system is completely submerged in a liquid of density ρ, the extension in the spring at equilibrium becomes $\frac{3x}{4}$. The density of the material of the object in terms of ρ will be (x is the initial extension of spring in equilibrium condition without immersing in liquid.)

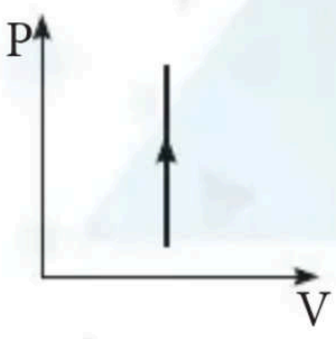
(1) 4ρ (2) 3ρ

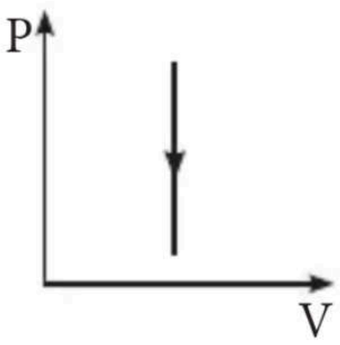
(3) 2ρ (4) 5ρ

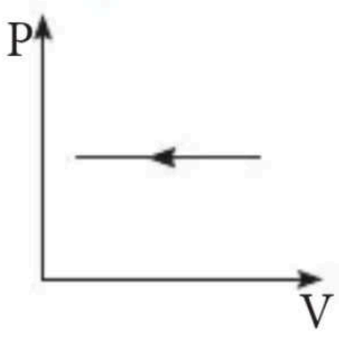
133. The given (P–U) graph shows how the internal energy of an ideal gas changes with increasing pressure. Which of the following pressure-volume (P–V) graphs corresponds to this relationship?



(1)


(2)


(3)


(4)


134. One mole of a monoatomic ideal gas is mixed with three moles of a diatomic ideal gas. If the molar specific heat of the mixture at constant volume is $\frac{\alpha^2 R}{4} J/mol-K$, then the value of α will be:

(Assume that the given diatomic gas has no vibrational mode.)

(1) 5 (2) 4

(3) 3 (4) 2

135. Match List I with List II.

List I (Application of Gauss Law)		**List II (Value of $\vec{E}$)**
A) Field inside thin shell	I)	$\frac{\lambda}{2\pi\varepsilon_0 r}\hat{r}$
B) Field outside thin shell	II)	$\frac{q}{4\pi\varepsilon_0 R^2}\hat{r}$
C) Field due to thin shell at its surface	III)	$\frac{q}{4\pi\varepsilon_0 r^2}\hat{r}$
D) Field due to long charged wire	IV)	Zero

In each case, the charges are uniformly distributed. (Here, symbols have their usual meaning, and R is the radius of thin shell.)

Choose the correct answer from the options given below:

(1) A-I, B-III, C-II, D-IV

(2) A-IV, B-III, C-I, D-II

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-III, C-II, D-I

CHEMISTRY

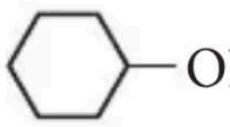
136. Match the List I with List II.

List I

A) $CH_3–O–CH_3$

B) $CH_3–CH_2–OH$

C) $C_6H_6 + CH_3COCl \xrightarrow[AlCl_3]{anhy}$

D)  $+ CH_3Br$

List II

I) Symmetrical ether

II) Gives ether on heating at 140 °C in presence of conc. H_2SO_4

III) Williamson's synthesis

IV) Friedel Crafts reaction

Select the correct option.

(1) A-IV, B-I, C-II, D-III

(2) A-IV, B-II, C-I, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-II, B-III, C-IV, D-I

137. Based on Bohr's theory, when 'n' value increases, choose the **incorrect** statements.

A. Velocity of electron increases.

B. Total energy of electron increases.

C. Radial distance between successive orbits decreases.

D. Energy difference between successive orbits decreases.

(1) A and D only

(2) B and C only

(3) B and D only

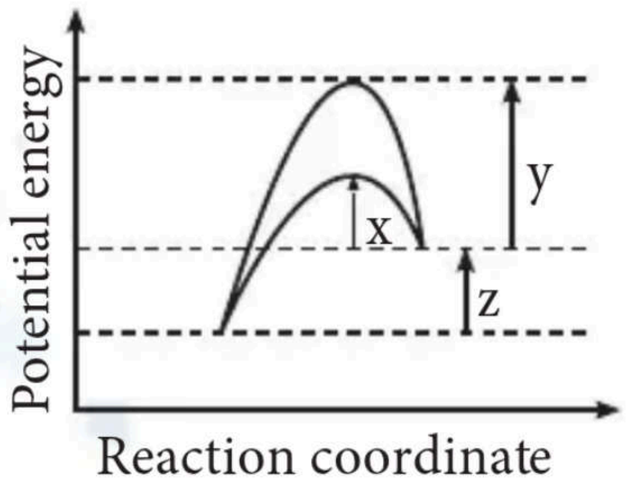
(4) A and C only

138. The magnitude of product of wavelength and velocity of a particle is 12.57. The product of uncertainty in velocity and uncertainty in position of the particle would be

(1) 10 (2) 4

(3) 2 (4) 1

139. The given graph shows the variation of potential energy for a reaction using a catalyst and for the same reaction without a catalyst.



Which of the following is a correct conclusion drawn from the above graph?

(1) ΔH in the presence of a catalyst, Z — E_a in the presence of the catalyst, Y + Z

(2) ΔH in the absence of a catalyst, Z — E_a in the presence of the catalyst, X + Z

(3) ΔH in the presence of a catalyst, Z — ΔH in the absence of the catalyst, X + Y

(4) ΔH in the presence of the catalyst, X+Z — E_a in the absence of the catalyst, X + Z

140. Which of the following processes do **not** involve absorption of energy?

A. $S_{(g)} + e^- \rightarrow S^-_{(g)}$

B. $O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)}$

C. $Cl_{(g)} + e^- \rightarrow Cl^-_{(g)}$

D. $O_{(g)} + e^- \rightarrow O^-_{(g)}$

(1) A, C, and D

(2) A, B, and D

(3) B, C, and D

(4) A, B, C, and D

141. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): Xe in XeF_2 is d^2sp^3 hybridised.

Reason (R): Central atom in XeF_2 molecule follows octet rule.

In light of the above statements, choose the most appropriate answer from the options given below.

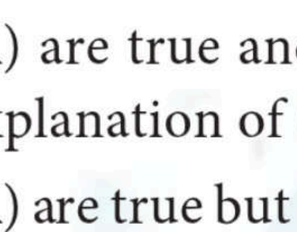
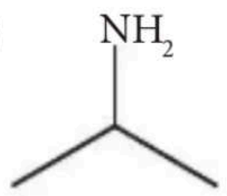
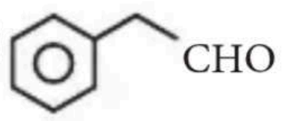
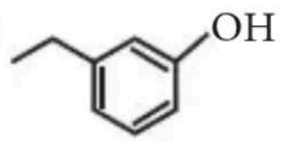
(1) Both(A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

142. Match List I with List II.

List I (Reagents used)	**List II (Compound with functional group)**
A) Alkaline solution of copper sulphate and sodium citrate	I) 
B) Neutral $FeCl_3$ solution	II) 
C) Alkaline chloroform solution	III) 
D) Sodium hypoiodite	IV) 

Select the correct option.

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-IV, C-III, D-I

(3) A-IV, B-I, C-II, D-III

(4) A-III, B-IV, C-II, D-I

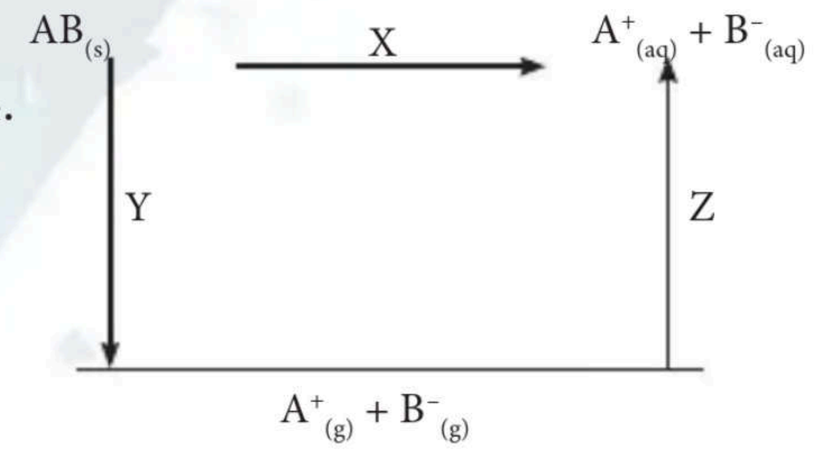
143. The lattice energies of KF, KCl, KBr and KI follow the order:

(1) KF > KCl > KBr > KI

(2) KI > KBr > KCl > KF

(3) KF > KCl > KI > KBr

(4) KI > KBr > KF > KCl

144. 

Identify X, Y, and *Z* in the above diagram. Consider AB as an ionic solid.

	X	**Y**	**Z**
(1)	$\Delta_{lattice} H°$	$\Delta_{hyd} H°$	$\Delta_{solution} H°$
(2)	$\Delta_{solution} H°$	$\Delta_{lattice} H°$	$\Delta_{hyd} H°$
(3)	$\Delta_{lattice} H°$	$\Delta_{solution} H°$	$\Delta_{hyd} H°$
(4)	$\Delta_{hyd} H°$	$\Delta_{lattice} H°$	$\Delta_{solution} H°$

145. How many of the following are primary amines?

A. tertiary butyl amine

B. secondary butyl amine

C. isobutyl amine

D. dimethyl amine

(1) 3 (2) 4

(3) 2 (4) 1

146. Consider the reaction:

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$, K_C for the reaction is 278 M^{-1}. 0.001 mole each of the reagents $SO_{2(g)}$, $O_{2(g)}$, and $SO_{3(g)}$ are mixed in a 1.0 L, flask. Determine the reaction quotient of the system and the direction of the equilibrium system.

(1) Qc = 1000; the equilibrium shifts to the product side.

(2) Qc = 1000; the equilibrium shifts to the reactant side.

(3) Qc = 0.001; the equilibrium shifts to the left.

(4) Qc = 0.001; the equilibrium shifts to the right.

147. When equal volume ____ and _____ are mixed, the precipitation of AgCl ($K_{sp} = 1.8 \times 10^{-10}$) will occur.

(1) 10^{-4} M $[Ag^+]$ and 10^{-4} M $[Cl^-]$

(2) 10^{-5} M $[Ag^+]$ and 10^{-5} M $[Cl^-]$

(3) 10^{-5} M $[Ag^+]$ and 10^{-6} M $[Cl^-]$

(4) 10^{-10} M $[Ag^+]$ and 10^{-10} M $[Cl^-]$

148. Which of the following gives a suffocating gas when treated with dilute HCl?

(1) Carbonate

(2) Sulphite

(3) Sulphate

(4) Borate

149. The correct order of melting points of IIIA group elements is

(1) B > Al > Tl > In > Ga

(2) B > Al > Ga > In > Tl

(3) B > Al > Tl > Ga > In

(4) B > Al > In > Tl > Ga

150. The maximum covalency exhibited by carbon and silicon respectively are

(1) 4, 6 (2) 4, 4

(3) 6, 6 (4) 4, 8

151. O–H bond polarity is maximum in _____.

(1) $(CH_3)_3$ C–OH

(2) CH_3–OH

(3) CH_3–CH(OH)CH_3

(4) CH_3–CH_2–OH

152. 0.5 g of organic compound in Kjeldhal method liberated ammonia which neutralised 60 mL of 0.1 N H_2SO_4 solution. The percentage of nitrogen in the compound is _____.

(1) 1.68 (2) 16.8

(3) 33.6 (4) 8.4

153. Benzene on nitration gives nitrobenzene in the presence of HNO_3 and H_2SO_4 mixture, where

(1) both H_2SO_4 and HNO_3 act as bases.

(2) both H_2SO_4 and HNO_3 act as acids.

(3) HNO_3 acts as a base and H_2SO_4 acts as an acid.

(4) HNO_3 acts as an acid and H_2SO_4 acts as a base.

154. Which one of the following fluorides **does not** exist?

(1) NF_5 (2) PF_5

(3) AsF_5 (4) SbF_5

Infinity Learn

155. Identify X in the following reaction.

$$CH_3(CH_2)_4CH_3 \xrightarrow[\text{Heat}]{\text{Anhy. } AlCl_3 \text{ / HCl}} X$$

(1) $(CH_3)_2CH(CH_2)_2CH_3$

(2) $Cl–CH_2–(CH_2)_4CH_2–Cl$

(3) C_6H_{12}

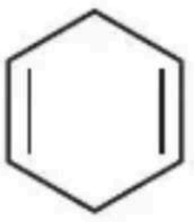
(4) $CH_3(CH_2)_4CH_2Cl$

156. An alkene on ozonolysis yields OHC-CH_2-CH_2- CH_2-CHO only. The alkene is _____.

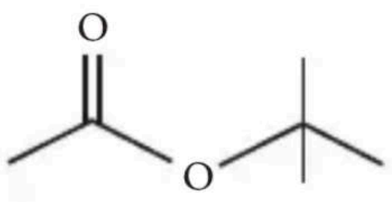
(1) $CH_2 = CHCH_2CH_2CH_2CH_2CH_3$

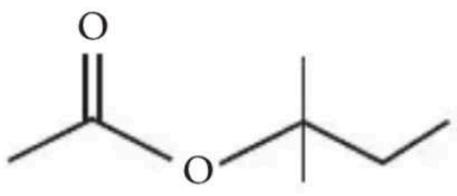
(2) 

(3) 

(4) 

157. An organic compound (A) of molecular formula $C_6H_{12}O_2$ was hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C' gives white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is _____.

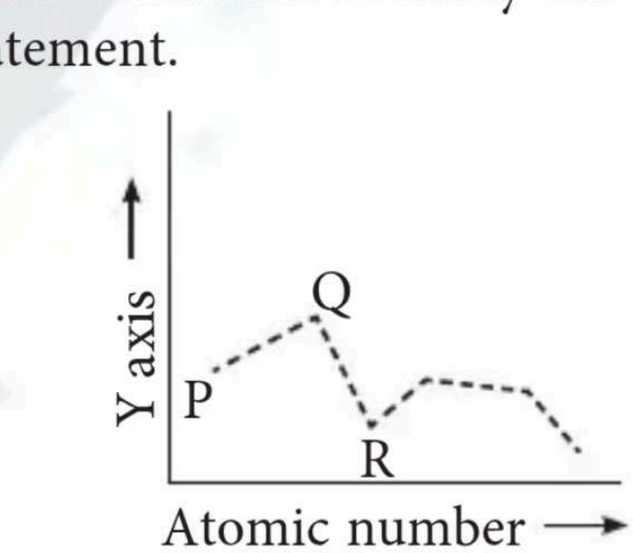
(1) 

(2) 

(3)

(4)

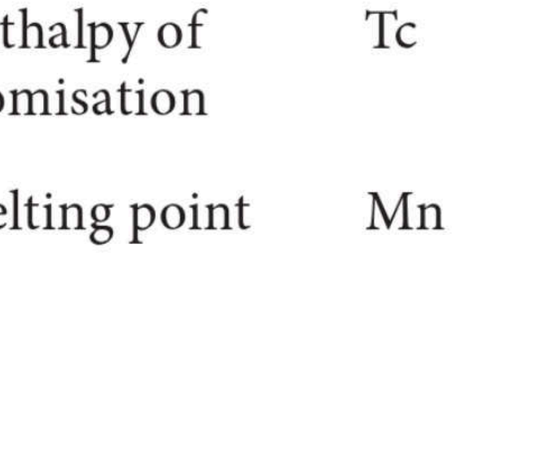
158. The following graph shows the solubility of gas (X) in water as a function of pressure at different temperatures of T_1 and T_2. What is the ratio of K_H(Henry's constant) at T_1 and T_2?



(1) 3 : 1 (2) 1 : 3

(3) $1:\sqrt{3}$ (4) $\sqrt{3}:1$

159. The given graph belongs to 3d series elements. Identify the correct statement.



	Y-axis	R
(1)	Enthalpy of atomisation	Cr
(2)	Melting point	Tc
(3)	Enthalpy of atomisation	Tc
(4)	Melting point	Mn

160. Two beakers X and Y contain CH_3COONa and CH_3COOH solutions respectively. When the same amount of water is added to both the beakers at the same temperature, which of the following changes in λ_m of the solution is correctly observed?

(1) It increases sharply in both the beakers.

(2) It increases slowly in both the beakers.

(3) In beaker X, it sharply increases, while in beaker Y, it slowly increases.

(4) In beaker X, it slowly increases, while in beaker Y, it sharply increases.

161. Which of the following orders is the **incorrect** representation of the property indicated by it?

(1) Br<Cl<F: electronegativity

(2) F<Br<Cl: electron affinity

(3) $F_2<Br_2<Cl_2$: bond energy

(4) $Br_2<Cl_2<F_2$: oxidising strength

162. For every 10 °C rise in temperature for most of the reactions, the rate gets doubled. For a zero order reaction, the temperature is increased from 35 °C to 65 °C, the rate of the reaction will become _____.

(1) zero

(2) infinity

(3) 8 times the initial rate

(4) the same as initial rate of reaction

163. A neutral heteroleptic complex of type $[MX_3Y_3]$ shows geometrical isomerism. In the given complex coordination number of central metal atom is 6 and its oxidation state is +3. Based on IUPAC nomenclature, which of the following is the correct statement?

(1) The denticity of each X and Y is 2.

(2) The charge on Y is zero and the charge on X is +1.

(3) The charge on both X and Y is zero.

(4) Both X and Y are monodentate ligands.

164. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): When $\Delta_0 < P$, 4th electron enters into t_{2g} orbital of d-subshell resulting in $t_{2g}^4 e_g^0$ configuration.

Reason (R): When $\Delta_0 < P$, CFSE value decreases; therefore, the electron easily enters into e_g orbital.

In light of the given statements, choose the correct answer from the options given below.

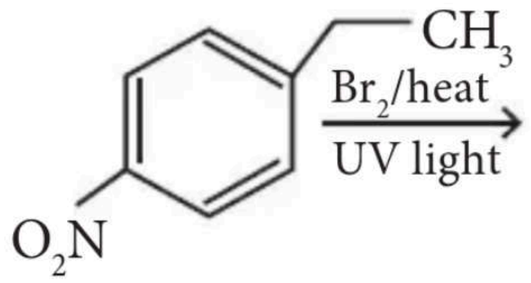
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

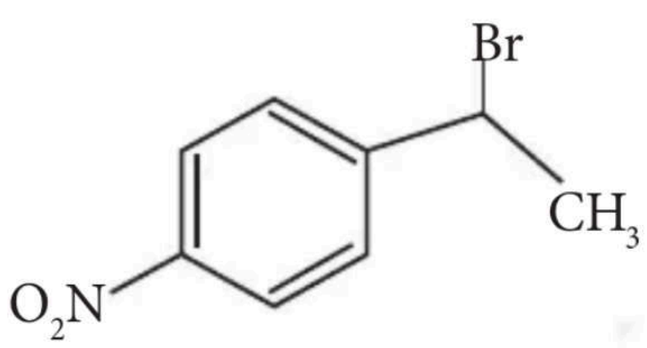
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

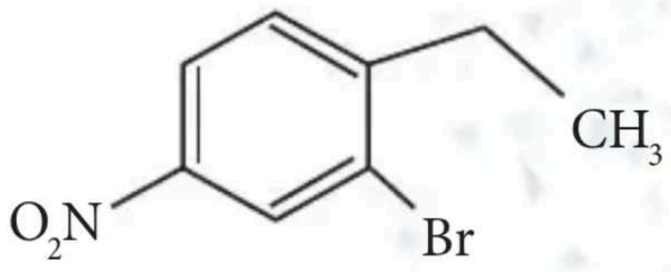
(3) (A) is true and (R) is false.

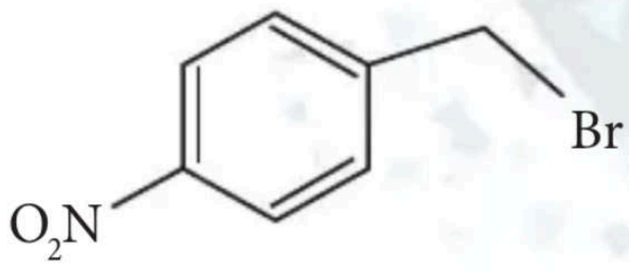
(4) (A) is false and (R) is true.

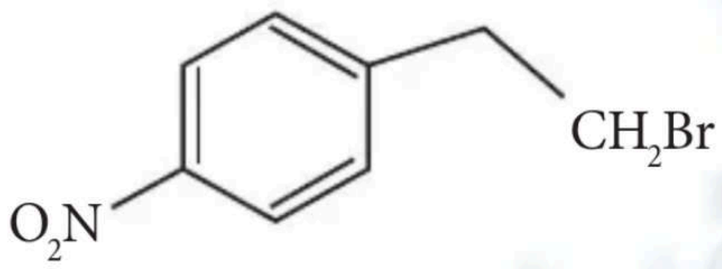
165. The major product formed in the following reaction is _____.



(1) 

(2) 

(3) 

(4) 

166. The IUPAC name of *neo*-pentane is _____.

(1) 2-Methylbutane

(2) 2, 2-Dimethylpropane

(3) 2-Methylpropane

(4) 2, 2-Dimethylbutane

167. A sample of calcium carbonate ($CaCO_3$) has the following percentage composition: Ca = 40%, C = 12%, and O = 48%. If the sample follows the law of constant proportions, the weight of calcium in 4 g of a sample of calcium carbonate from another source will be ___.

(1) 0.016 g (2) 0.16 g

(3) 1.6 g (4) 16 g

168. What is 'X' in the following reaction?

$$C_6H_5 - C \equiv C - H \xrightarrow[\text{dil .}H_2SO_4]{Hg^{+2}} X$$

(1) C_6H_5–CO–CH_3

(2) C_6H_5–CH_2–CHO

(3) C_6H_5–CHO

(4) C_6H_5–CH_2–OH

169. Identify the correct statements.

A. Formaldehyde is a liquid.

B. Acetaldehyde is soluble in water.

C. The boiling point of ketones is greater than the corresponding aldehydes.

D. Lower aldehydes have pungent smell.

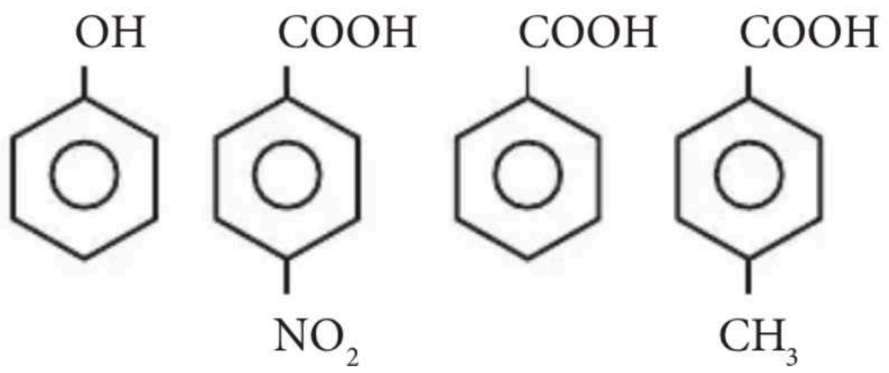
(1) A, B, and D only

(2) B and D only

(3) A and C only

(4) B, C, and D only

170. The correct order of acid character of the following compounds is



(I) (II) (III) (IV)

(1) III > II >I >IV

(2) IV >III >II >I

(3) I >II >III >IV

(4) II > III >IV >I

171. Given below are two statements.

Statement I : Formaldehyde is more reactive than benzaldehyde towards the nucleophilic addition reaction.

Statement II : Formaldehyde is more volatile than benzaldehyde

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

172. Given below are two statements.

Statement I : Standard molar enthalpy of formation of $H_{2(g)}$ is zero.

Statement II : For gaseous diatomic molecules, the enthalpy of atomisation is also called the bond dissociation enthalpy.

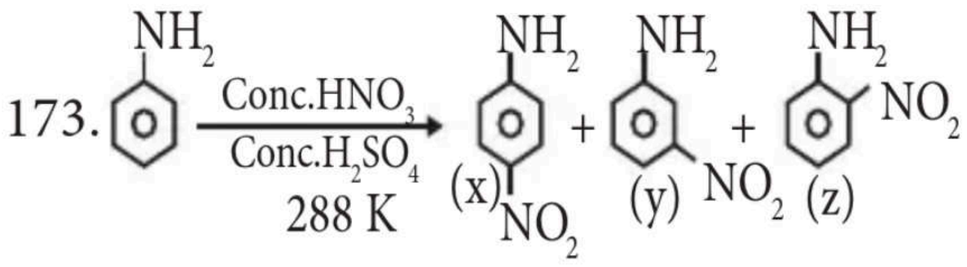
In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

173. 

Determine the percentages of x, y, and z in the above reaction (reaction occurs at 288 K).

(1) 2%, 36%, 72%

(2) 51%, 47%, 2%

(3) 38%, 34%, 28%

(4) 71%, 27%, 2%

174. The bond order and magnetic behaviour of O_2^{-2} ion are

(1) 1.5 and paramagnetic

(2) 1.5 and diamagnetic

(3) 2 and diamagnetic

(4) 1 and diamagnetic

175. The Glycosidic linkage present in sucrose is between _____ and _____.

(1) C-1 of β-glucose and C-4 of α-glucose

(2) C-1 of α-glucose and C-4 of β-glucose

(3) C-1 of α-glucose and C-4 of α-fructose

(4) C-1 of α-D glucose and C-2 of β-D fructose

176. Non-essential amino acids among the following are

A. Tyrosine

B. Tryptophan

C. Alanine

D. Methionine

E. Glutamic acid

(1) B, C, and D only

(2) A, B, C, and E only

(3) A, C, and E only

(4) B and C only

177. Both DNA and RNA contain four bases each. Which of the following bases is **not** present in RNA?

(1) Adenine (2) Uracil

(3) Thymine (4) Cytosine

178. The electronegativity difference between N and F is greater than the electronegativity difference between N and H. Yet, the dipole moment of NH_3(4.90×10^{-30} C m) is greater than NF_3(0.8×10^{-30} C m). This is because

(1) in NH_3, the orbital dipole and resultant dipole are in the same direction, whereas in NF_3, these are in opposite directions.

(2) in NH_3 as well as NF_3, the orbital dipole and resultant dipole are in opposite directions.

(3) in NH_3, the orbital dipole and resultant dipole are in opposite directions, whereas in NF_3, these are in the same direction.

(4) in NH_3 as well as in NF_3, the orbital dipole and resultant dipole are in the same direction.

179. (X) is a non-metal present in cryolite, and (Y) is a non-metal present in fluoroapatite. Identify (X) and (Y) from the following.

(1) X = F and Y = P

(2) X = Na and Y = F

(3) X = Na and Y = Ca

(4) X = Al and Y = F

180. In the given reaction, the number of H^+ ions involved in the balanced equation is _____.

$$MnO_4^- + SO_3^{2-} + H^+ \rightarrow Mn^{2+} + SO_4^{2-} + H_2O$$

(1) 2

(2) 6

(3) 8

(4) 16